

ADITYA MISRA

Phone: +65 8381 5159 | Email: aditya.misra@u.nus.edu

LinkedIn: www.linkedin.com/in/adityamisra04 | Portfolio: <https://adityamisra.vercel.app/>

AI Engineer building scalable, user-centric production-grade GenAI systems, agentic workflows, and data products.

EDUCATION

National University of Singapore

Aug 2022 - May 2026

Bachelor of Computing (Hons.) in Computer Science, Minor in Innovation and Design

Singapore

- GPA: 4.6/5.0; Focus Areas: Database Systems, AI & Machine Learning
- Relevant Coursework: Machine Learning Systems, Natural Language Processing, Big Data Systems
- Awards: Top Student in Foundations of Software Engineering, Big Data Systems, and Database Systems
- Teaching Assistant: AI & Machine Learning, Foundations of Software Engineering

SKILLS

Languages

Python, SQL, Java, JavaScript, TypeScript, C

AI/ML

PyTorch, HuggingFace, scikit-learn, pandas, LangChain

Data & Systems

FastAPI, Docker, AWS, PySpark, PostgreSQL, MongoDB, Pinecone

EXPERIENCE

AI Engineer Intern

Jul 2025 - Dec 2025

AI Singapore

Singapore

- Engineered a scalable ETL pipeline to ingest and process 15,000+ hours of multilingual speech datasets for SEA-LION, Singapore's national LLM, optimising throughput across ingestion, transformation, and validation stages.
- Designed an ASR evaluation framework computing WER/CER with per-language, accent, and noise-condition breakdowns, visualizing performance distributions and benchmarking against baseline models to track regressions and gains.

AI Engineer Intern

May 2025 - Jul 2025

Venture Corporation Limited

Singapore

- Led end-to-end development and delivery of *w[AI]ter*, an edge-based VoiceAI dining assistant, from architectural design to market validation, logging 40,000+ interactions and achieving a Net Promoter Score of 40 from pilot users.
- Architected a state-aware agentic workflow using a finite state controller to orchestrate a Phi-3-Mini SLM with reliable tool usage and control flow across the ASR and hybrid collaborative and content-based recommendation engine.
- Built a low-latency edge inference pipeline on NVIDIA Jetson Orin NX, utilising ONNX Runtime and model quantisation to optimise throughput for the ASR and LLM components in a resource-constrained environment.

Software Engineer Intern

May 2024 - Nov 2024

Hutchinson Research and Innovation

Singapore

- Constructed a real-time data dashboard with Angular for monitoring 120+ machine parameters, streaming on-site sensor data through Redis into PostgreSQL with a FastAPI layer, enabling rapid fault detection and operational visibility.
- Designed and implemented a full-stack RFQ document management system with React, FastAPI, and PostgreSQL, engineering a multi-level approval workflow with a complete audit trail, reducing approval cycle times by 50%.

Software Engineer Intern

Oct 2023 - Jan 2024

DEIF Asia Pacific

Singapore

- Implemented core Open Charge Point Protocol 2.0.1 workflows in Java, handling asynchronous message exchanges and charger lifecycle events to improve interoperability of EV chargers, reducing integration costs by 45%.
- Developed a React-based simulator for the Central Charging Management System to exercise 40+ end-to-end charging scenarios, enabling protocol validation, failure-case testing, and regression checks across charger implementations.

PROJECTS

ESG Edge, Nomura AI Challenge – Finalist

- Built a RAG-driven ESG insights platform on Bedrock, ingesting multi-source data via AWS Lambda, with hierarchical chunking and vector retrieval to power a context-aware chatbot, projected to save 100,000+ analyst hours annually.

UrbanCIA, Final Year Project [Supervised by Prof. Roger Zimmermann]

- Building a LangChain-powered multi-agent system that autonomously selects causal inference methods, validates identification assumptions, and generates interpretable causal effect estimates for empirical urban policy analysis.

HalalChain, NUS FinTech Summit Hackathon 2026 – Winner

- Built the backend of a Shariah-compliant tokenization and trading platform on XRPL, defining Next.js API routes and PostgreSQL schemas to manage full asset lifecycles from token issuance and DEX trading to maturity redemption.